DEEP DIVE · REFERENCE · PRINT-READY

UK Radio Frequency Allocations

Primary source: Ofcom UK Frequency Allocation Table (UKFAT) · Mode subdivisions: RSGB Band Plan 2026 · Compiled 4 Nov 2026 · Antenna lengths computed at band centre

Every frequency band allocated in the UK, what it's used for, how the spectrum is divided between modes (CW, SSB, FM, digital protocols), and the corresponding antenna lengths (half-wave and quarter-wave dipole). The **Useful Frequencies** section covers the frequencies you actually want to listen to — ADS-B, GPS, weather, aviation, DMR, APRS, FT8, PSK31, RTTY, and UK broadcast.

How to read the antenna columns: λ/2 = half-wave dipole total length (two equal elements, 180° apart, fed at centre). λ/4 = quarter-wave vertical monopole height (over ground plane) — the most common practical antenna. Values are for free space at the stated frequency; real-world shortening factor ≈ 95% for wire dipoles. For VHF+ FM work, a J-pole or 5/8-wave vertical is the usual recommendation over a bare λ/4.

01 Full UK Allocation Table — LF through UHF

Allocations verified against the Ofcom UKFAT. Mode subdivisions from the RSGB Band Plan 2026 (IARU Region 1).

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	λ/2 DIPOLE	λ/4	NOTES
148.5–255 kHz	**LF**	Longwave — BBC Radio 4 (225 kHz), time signals	`AM (DSB), 9–10 kHz ch` Groundwave only, very local — needs a large loop antenna.	681.3 m	340.7 m	λ/2 ≈ 679 m — impractical; use loop or long wire
526.5–1605 kHz	**MF — AM radio**	UK domestic AM radio (BBC Local, commercial) + BBC World Service (long-range at night)	`AM (DSB), 10 kHz ch` Day: short range · Night: skywave over hundreds of km — the classic 'whisper' band.	149.9 m	74.9 m	Long wire/vertical; λ/4 at 1 MHz = 74.95 m

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	Λ/2 DIPOLE	Λ/4	NOTES
1.6–2.0 MHz	**160m**	Amateur — long-range, night, DX	`CW (1810–1838), narrowband (1838–40), all-modes (1840–43), LSB voice (1843–2000)` • **CW** 1810–1838 kHz · **FT8/WSPR** 1838–40 · **PSK31** 1840–43 • **SSB LSB** voice 1843–2000 kHz (dial 1843 min) · QRP 1836 · DF beacons 1960	79.7 m	39.9 m	32 W max (UK); primary to foreign stations below 1850
3.5–3.8 MHz	**80m**	Amateur — workhorse DX band, night	`CW (3500–570), narrowband (3570–590), LSB voice (3600–800)` • **CW** 3500–3570 (contests 3510–555) · **FT8** 3573–75 · **PSK31** 3574 · **RTTY** 3591 (R1) • **SSB LSB** 3600–3800 (dial 3603 min) · SSB QRP 3690 · UK emergency 3663 · R1 emergency 3760	42.0 m	21.0 m	Primary; shared with other services
5.25–5.45 MHz	**60m**	Amateur — narrowband channels, Full licence only	`USB voice in 12 defined channels (all-modes per channel)` • **USB recommended** (not LSB) · CW QRP 5262 · UK emergency 5278.5 · beacon 5290 · AM 6 kHz 5317 • WSPR 5362–5374.5 · channels 5258.5 / 5276 / 5288.5 / 5298 / 5313 / 5333 / 5354 / 5362 / 5378 / 5395 / 5403.5	28.3 m	14.1 m	Full licence only, 100 W max, specific channel plan

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	Λ/2 DIPOLE	Λ/4	NOTES
7.0–7.2 MHz	**40m**	Amateur — most popular UK band, day & night DX	`CW (7000–40), narrowband (7040–50), LSB voice (7050–200)` • **CW** 7000–7040 · **FT8** 7074–76 · **PSK31** 7031 · **RTTY** 7091 (R1) • **SSB LSB** 7053–7200 (dial 7053 min) · SSB QRP 7090 · UK emergency 7110 · SSTV 7175	21.2 m	10.6 m	Primary
10.1–10.15 MHz	**30m**	Amateur — narrowband only (secondary)	`CW (10100–130) · narrowband (10130–150)` • **FT8** 10.136 · **PSK31** 10.137 • No SSB except life-safety emergencies; no contests	14.8 m	740 cm	—
14.0–14.35 MHz	**20m**	Amateur — daytime DX, 'the' band	`CW (14000–070), narrowband (14070–89), USB voice (14112–350)` • **CW** 14000–14070 · **FT8** 14074 · **PSK31** 14031 · **RTTY** 14095 (R1) · beacon 14100 • **SSB USB** 14112–14350 (contests 14125–300) · SSB QRP 14285 · SSTV 14230 · global emergency 14300	10.5 m	527 cm	Primary (satellite primary below 14250)
18.068–18.168 MHz	**17m**	Amateur — DX, less crowded	`CW (18068–095), narrowband (18095–109), USB voice (18111–168)` • **CW** 18068–18095 · **FT8** 18100 · **PSK31** 18100 · **RTTY** 18110 (R1) · beacon 18110 • **SSB USB** 18111–18168 · SSB QRP 18130 · SSTV 18165	827 cm	414 cm	Primary

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	λ/2 DIPOLE	λ/4	NOTES
21.0–21.45 MHz	**15m**	Amateur — great summer daytime DX	`CW (21000–070), narrowband (21070–149), USB voice (21151–450)` • **CW** 21000–21070 · **FT8** 21074–76 · **PSK31** 21316 · **RTTY** 21095 (R1) · beacon 21150 • **SSB USB** 21151–21450 · SSB QRP 21285 · SSTV 21340 · global emergency 21360 · sat 21125–245	704 cm	352 cm	Primary
24.89–24.99 MHz	**12m**	Amateur — narrowband only, no contests	`CW (24890–915), narrowband (24915–929), USB voice (24931–990)` • **FT8** 24916 · **PSK31** 24916 · **RTTY** 24920 (R1) · beacon 24930 • **SSB USB** 24931–24990 · SSB QRP 24950	601 cm	301 cm	Primary
28.0–28.5 MHz	**10m**	Amateur — summer DX, sporadic-E	`CW (28000–070), narrowband (28070–190), USB voice (28190–500), FM (28.3–4 MHz)` • **CW** 28000–28070 · **FT8** 28074–76 · **PSK31** 28125 · **RTTY** 28120 (R1) · beacon 28190 • **SSB USB** 28190–28500 · SSB QRP 28330 · SSTV 28220 · **FM voice** 28.30–28.40 MHz (50 kHz ch)	532 cm	266 cm	Primary; 30 W for Intermediate

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	Λ/2 DIPOLE	Λ/4	NOTES
50–52 MHz	**6m**	Amateur — tropospheric ducting DX in summer	`CW (50.0–01) · SSB USB (50.1–04) · FM (50.7+)` • **CW** 50.000–50.100 (beacons 50.000–030) · **SSB** 50.100–50.300 (intl CoA 50.150) · MGM 50.305 • Beacons 50.400–50.500 (no TX) · **FM voice** 51.410–51.590 (calling **51.510**) · repeaters 51.220–51.370 in / 50.720–50.870 out	298 cm	149 cm	Primary · FM: J-pole ≈ 4.1 m or 5/8-wave ≈ 3.7 m (preferred over λ/4)
70–70.5 MHz	**4m**	Amateur — NoV band (Full licence)	`Narrowband (70.1–25) · 12.5 kHz ch all-modes (70.294–500)` • Beacons 70.000–70.090 · **CW/SSB centre** 70.200 · **AM/FM calling** 70.260 · MGM 70.270 • **FM** 70.294–70.500 (2.5 kHz steps): internet gateways 70.350/362.5/387.5/412.5 · **FM calling** 70.450 · GB2RS 70.425	213 cm	107 cm	Secondary; NoV required, 160 W max · FM: J-pole ≈ 2.9 m or 5/8 ≈ 2.7 m (preferred over λ/4)

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	Λ/2 DIPOLE	Λ/4	NOTES
144–146 MHz	**2m**	Amateur VHF — FM voice, SSB, digital	**FM voice** (145.0–07) · **SSB** (144.15–40) · CW (144.0–15) • **CW** 144.025–144.150 (CoA 144.050) · **SSB USB** 144.150–144.400 (CoA 144.300, GB2RS 144.250) • **FM** simplex 145.200–145.5935 (calling **145.500**) · **APRS** 145.525 (AX.25 1200 bd AFSK) • **DMR** 145.275 · **C4FM** 145.2375 · repeater inputs 145.100–145.500 / outputs 145.500–146.000 (8 MHz split)	103 cm	51.5 cm	Primary; 400 W ERP · FM: J-pole ≈ 1.4 m or 5/8 ≈ 1.3 m (preferred over λ/4); 1/2-wave collinear for mobile
430–440 MHz	**70cm**	Amateur UHF — FM, DV, digital, EME	**FM/DV** (430.1–431.0) · **SSB/CW** (431.0–432.4) · FM (433.0+) • **CW/SSB** 431.000–432.400 (2.7 kHz) · **FM voice** 433.000–433.500 (calling **433.500**) · **DMR** 433.500 • Repeater outputs 438.000–439.000 / inputs 430.000–431.000 (7.6 MHz split) · sat uplinks 435–438 · EME CW 432.0–432.1	34.3 cm	17.1 cm	Primary; 400 W ERP; conditions in 430–432 MHz · FM: J-pole ≈ 47 cm or 5/8 ≈ 43 cm (preferred over λ/4)

Sideband rule of thumb: below 10 MHz use **LSB**, above 10 MHz use **USB**. Minimum dial settings: 1843 kHz (160 m), 3603 kHz (80 m), 7053 kHz (40 m).

02 Useful Frequencies — Listen / Receive

The frequencies worth tuning to, with mode and antenna. Sorted by service, not frequency.

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	Λ/2 DIPOLE	Λ/4	NOTES
1090 MHz	**ADS-B (Mode S)**	Aircraft position/altitude/speed — Mode S 1090ES, one receiver does everything	`Pulse 1 Mbaud, 8-bit`	13.8 cm	6.88 cm	Small V-band patch or 5/8 whip; λ/2 = 13.6 cm
156.050 MHz	**Ch 16 — DSC**	Marine VHF distress & calling (digital)	`DSC digital + GMDSS voice`	96.1 cm	48.0 cm	Whip λ/4 ≈ 47.6 cm (mobile marine mast)
156.525 MHz	**Ch 85B — DSC**	Marine DSC digital distress/calling	`DSC digital`	95.8 cm	47.9 cm	—
121.5 MHz	**Aviation emergency**	Guard frequency — aircraft emergency (ELT)	`AM voice, 25 kHz ch`	123 cm	61.7 cm	—
122.900 MHz	**Air Traffic (UK)**	UK ATC control / en-route (example)	`AM voice, 25 kHz ch`	122 cm	61.0 cm	Aviation band 118–137 MHz — AM, not FM
108–117.975 MHz	**VOR / ILS glideslope**	VOR navigation + ILS glideslope (UK)	`AM / FM 30 Hz/90 Hz`	134 cm	66.9 cm	Sits inside the FM broadcast band — VOR/ILS on centre kHz
108–118 MHz	**ILS localiser**	Instrument landing — localiser (within VOR band)	`AM, 100 Hz mod`	136 cm	68.1 cm	—
243.0 MHz	**Aviation emergency**	Guard frequency — aircraft emergency (ELT, 2m satellite link)	`AM voice, 25 kHz ch`	61.7 cm	30.8 cm	—
1027–1150 MHz	**DME**	Distance Measuring Equipment — paired with VOR/ILS (channels 1–126)	`UHF pulse pairs, 63.5 µs`	13.8 cm	6.88 cm	λ/2 ≈ 13.6 cm (1 GHz) — whip or patch

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	Λ/2 DIPOLE	Λ/4	NOTES
1575.42 MHz	**GPS L1 C/A**	Civil GPS positioning	`CDMA, 1.023 Mcps`	9.51 cm	4.76 cm	Patch antenna (V-band) — λ/2 ≈ 9.5 cm
1227.60 MHz	**GPS L2**	GPS L2 (military L2C + civil)	`CDMA`	12.2 cm	6.11 cm	—
1176.45 MHz	**GPS L5**	GPS L5 — improved accuracy	`CDMA`	12.7 cm	6.37 cm	—
1191.795 MHz	**Galileo E1**	EU GNSS (shared L-band)	`CDMA`	12.6 cm	6.29 cm	—
1207.14 MHz	**Galileo E5a**	EU GNSS — safety-of-life	`CDMA`	12.4 cm	6.21 cm	—
1544.43 MHz	**Galileo E5b/E6**	EU GNSS E5b (PPP) / E6	`CDMA`	9.71 cm	4.85 cm	—
1598–1605.89 MHz	**GPS L1C (new)**	Next-gen civilian L1C	`CDMA`	9.36 cm	4.68 cm	—
1602 MHz	**GLONASS G1**	Russian GNSS (L1, 1602–1605.4 MHz)	`CDMA/FDMA`	9.35 cm	4.68 cm	—
403–405 MHz	**Weather — MAS band**	Meteorological Aids Service band (weather datacasting, aeronautical METAR feeds)	`Digital subcarrier / FSK`	37.1 cm	18.6 cm	λ/4 ≈ 18.6 cm whip
162.400–162.550 MHz	**Weather — US NOAA**	US NWS weather radio (WXR) — 7 channels	`AM + subcarrier FSK (1785 Hz)`	92.3 cm	46.1 cm	—
145.275 MHz	**DMR (2m)**	Digital Mobile Radio voice — BrandMeister / DMRplus	`DMR Tier 2, 4.5 kHz`	103 cm	51.6 cm	—
433.500 MHz	**DMR (70cm)**	DMR voice — UK hotspots	`DMR Tier 2`	34.6 cm	17.3 cm	—
145.2375 MHz	**C4FM (2m)**	C4FM digital voice (Yaesu Fusion, Icom IDAS)	`C4FM 6.25 kHz`	103 cm	51.6 cm	—

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	λ/2 DIPOLE	λ/4	NOTES
145.525 MHz	**APRS**	Amateur Packet Radio — position/status (AX.25 AFSK 1200 bd)	`AX.25 / APRS`	103 cm	51.5 cm	—
14.074 MHz	**FT8 (20m)**	Most popular HF digimode — weak-signal digital	`FT8 (JT9-based, 6 FSK)`	10.7 m	533 cm	—
7.074 MHz	**FT8 (40m)**	FT8 — 40 m	`FT8`	21.2 m	10.6 m	—
7.031 MHz	**PSK31 (40m)**	Slow PSK31 data	`PSK31, 31 Bd`	21.3 m	10.7 m	—
14.130 MHz	**PSK31 (20m)**	PSK31 — 20 m	`PSK31`	10.6 m	530 cm	—
7.091 MHz	**RTTY (40m)**	RTTY (R1: TX 7.091 MHz, shift 170 Hz)	`FSK 170 Hz, 45 Bd`	21.1 m	10.6 m	—
14.095 MHz	**RTTY (20m)**	RTTY — 20 m	`FSK 170 Hz`	10.6 m	532 cm	—
14.100 MHz	**Beacon (20m)**	IARU propagation beacon (no TX!)	`CW 75 wpm`	10.6 m	532 cm	—
28.190 MHz	**Beacon (10m)**	Propagation beacon (no TX!)	`CW`	532 cm	266 cm	—
145.500 MHz	**FM calling (2m)**	VHF FM voice calling / simple channel	`FM 12.5 kHz`	103 cm	51.5 cm	—
433.500 MHz	**FM calling (70cm)**	UHF FM voice calling	`FM 12.5 kHz`	34.6 cm	17.3 cm	—
51.510 MHz	**FM calling (6m)**	6 m FM voice calling	`FM 12.5 kHz`	291 cm	146 cm	—
70.260 MHz	**AM/FM calling (4m)**	4 m AM/FM calling	`AM/FM`	213 cm	107 cm	—
526.5–1605 kHz	**AM radio (UK)**	BBC Local + commercial AM	`AM (DSB) 10 kHz`	149.9 m	74.9 m	—

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	Λ/2 DIPOLE	Λ/4	NOTES
87.5–108 MHz	**FM radio (UK)**	UK FM broadcasting (stereo)	`FM stereo 19 kHz`	150 cm	74.9 cm	λ/2 ≈ 1.49 m, λ/4 ≈ 0.75 m (whip)
174–240 MHz	**DAB / DAB+ (UK)**	UK digital radio — Band III	`COFDM, 1.536 Mcs`	74.6 cm	37.3 cm	UK: ch 5A BBC, 6A BBC, 7A BBC, 8A/10A/11A commercial — typical 201 MHz
470–694 MHz	**UHF TV (UK)**	UK Freeview — ch 21–60 (DVB-T2)	`DVB-T2 8k, ~7 MHz`	27.3 cm	13.6 cm	Grouped UHF log-periodic or Yagi; λ/4 at 550 MHz ≈ 13.6 cm
470–790 MHz	**4G / 5G (UK)**	UK mobile: 4G 800/1800/2600 MHz, 5G 3.4–3.8 GHz + 26 GHz	`OFDMA (LTE/5G NR)`	18.7 cm	9.37 cm	—
880–915 MHz	**GSM 900 (UK)**	UK 2G/3G/4G — 900 MHz block	`GSM / WCDMA / LTE`	16.8 cm	8.42 cm	—
1805–1880 MHz	**1800 MHz (UK)**	UK 2G/3G/4G — 1.8 GHz block	`GSM / WCDMA / LTE`	8.15 cm	4.07 cm	—
2100–2200 MHz	**2100 MHz (UK)**	UK 3G/4G — 2.1 GHz block	`WCDMA / LTE`	6.97 cm	3.49 cm	—
2655–2690 MHz	**2600 MHz (UK)**	UK 4G/5G — 2.6 GHz block	`LTE / 5G NR`	5.61 cm	2.81 cm	—
3400–3800 MHz	**5G (UK)**	UK 5G mid-band (C-band)	`5G NR`	4.28 cm	2.14 cm	λ/2 ≈ 4.28 cm — patch/panel antenna

FREQUENCY	BAND / NAME	USE / ALLOCATION	MODE / PROTOCOL	λ/2 DIPOLE	λ/4	NOTES
11.7–12.2 GHz	**Ku-band SAT TV**	Sat TV (SES-2, etc.) — downlink 10.7–12.75 GHz	`QPSK/8PSK DVB-S2`	1.26 cm	0.63 cm	Offset-fed dish (60–90 cm); feed LNB ≈ 2.5 cm waveguide
4.7–6.9 GHz	**C-band**	Satcom / backhaul (uplink 5.925–6.425)	`QPSK`	2.50 cm	1.25 cm	—

03 Known Frequencies Worth Tuning

Real, well-known frequencies across the bands for particular purposes — weather, number stations, time signals, emergency/distress, and international broadcasting. These are for **listening** (SDR or scanner) unless noted as ham emergency CoAs. Emergency frequencies are for receiving/monitoring; never transmit on them outside a genuine emergency.

Weather stations (NWS HF, worldwide SDR listening)

US National Weather Service HF weather — AM voice + digital subcarrier, receivable worldwide. The classic "weather station" band. UK has no equivalent HF service; the UK Met Office weather is on FM/TV/web.

FREQUENCY	NAME / CALLSIGN	MODE	WHAT IT IS
4125 kHz	**4125 kHz — NWS HF WX**	`AM + subcarrier`	NWS HF weather (also the 4 MHz HF maritime distress voice freq)
5850 kHz	**5850 kHz — NWS HF WX**	`AM + subcarrier`	NWS HF weather, 6 m
6315 kHz	**6315 kHz — NWS HF WX**	`AM + subcarrier`	NWS HF weather, 6 m
8425 kHz	**8425 kHz — NWS HF WX**	`AM + subcarrier`	NWS HF weather, 8 m
9200 kHz	**9200 kHz — NWS HF WX**	`AM + subcarrier`	NWS HF weather, 9 m
12630 kHz	**12630 kHz — NWS HF WX**	`AM + subcarrier`	NWS HF weather, 13 m
14640 kHz	**14640 kHz — NWS HF WX**	`AM + subcarrier`	NWS HF weather, 20 m

Number stations (spy / mystery shortwave)

One-way coded transmissions, usually synthetic voice reading groups of five digits, often introduced by a distinctive tune. No published schedules; frequencies and times shift without notice. From the 2026 B-26 EIBI listener-log schedule — treat as a guide, not gospel.

FREQUENCY	NAME / CALLSIGN	MODE	WHAT IT IS
3168 kHz	**Channel Marker L**	`CW (Morse)`	Continuous, 24 h — Russia
3243 kHz	**Goose Channel Marker**	`Voice / various`	Nightly UTC, Russia
3364 kHz	**The Squeaky Wheel**	`Voice / various`	Nightly UTC, Russia
3850 kHz	**Baron Channel Marker**	`Voice / various`	Nightly UTC, Russia
4770 kHz	**Alarm Channel Marker**	`Voice / various`	Continuous, Russia
4909 kHz	**E11 (Polish spy numbers)**	`Synthetic voice`	Multiple UTC slots, Poland
5082 kHz	**E11 (Polish spy numbers)**	`Synthetic voice`	Evening UTC, Poland
5943 kHz	**S06 (Russian spy numbers)**	`Synthetic voice`	Evening UTC, Russia
6804 kHz	**E11 (Polish spy numbers)**	`Synthetic voice`	Morning UTC, Poland
6849 kHz	**E11 (Polish spy numbers)**	`Synthetic voice`	Evening UTC, Poland

Time & frequency standard signals

National time/frequency beacons — great for checking a receiver against the standard. WWVB/WWV are NIST (US).

FREQUENCY	NAME / CALLSIGN	MODE	WHAT IT IS
60 kHz	**WWVB (NIST)**	`CW + digital code`	LF, US radio clocks — needs a loop antenna
100 kHz	**WWV (NIST)**	`CW + digital code`	LF time/frequency standard
2.5 MHz	**WWV / WWVH**	`CW + digital code`	HF time code, NIST
5 MHz	**WWV / WWVH**	`CW + digital code`	HF time code, NIST
10 MHz	**WWV / WWVH**	`CW + digital code`	HF time code, NIST
15 MHz	**WWV / WWVH**	`CW + digital code`	HF time code, NIST
20 MHz	**WWV**	`CW + digital code`	HF time code (WWV only)

Emergency, distress & safety (GMDSS + aviation)

International distress and safety frequencies — these are the ones a scanner/SDR should always keep programmed. Do NOT transmit on these unless in an actual emergency.

FREQUENCY	NAME / CALLSIGN	MODE	WHAT IT IS
2182 kHz	**MF distress (radiotelephony)**	`AM voice`	International MF distress; hourly 3-min silence periods
4125 kHz	**4 MHz HF distress voice**	`AM voice`	HF maritime distress (also NWS weather)
4207.5 kHz	**4 MHz HF DSC**	`Digital`	Maritime digital selective calling
518 kHz	**NAVTEX**	`Digital FSK`	International maritime safety information (primary)
490 kHz	**NAVTEX (regional)**	`Digital FSK`	Regional NAVTEX
6215 kHz	**6 MHz HF distress voice**	`AM voice`	HF maritime distress
8291 kHz	**8 MHz HF distress voice**	`AM voice`	HF maritime distress
12290 kHz	**12 MHz HF distress voice**	`AM voice`	HF maritime distress
16420 kHz	**16 MHz HF distress voice**	`AM voice`	HF maritime distress
121.5 MHz	**Aviation emergency / EPIRB homing**	`AM`	Guard freq — aircraft distress; EPIRB locator beacon
243.0 MHz	**Aviation emergency (satellite)**	`AM`	Aircraft emergency / ELT via 2 m satellite
406.0–406.1 MHz	**EPIRB (Cospas-Sarsat)**	`Digital`	Satellite emergency beacon — position relay
156.800 MHz	**Ch 16 — marine VHF distress**	`AM voice / DSC`	Continuous maritime listening watch
156.525 MHz	**Ch 70 — marine DSC**	`Digital`	Maritime digital distress, no voice
9200–9500 MHz	**SART (X-band)**	`Radar transponder`	Search & Rescue transponder — 12 dots on radar

UK / R1 amateur emergency centres (for ham operators)

Designated emergency CoAs from the RSGB band plan — where hams go to handle emergency traffic. These sit inside the ham allocations already listed in section 1.

FREQUENCY	NAME / CALLSIGN	MODE	WHAT IT IS
5278.5 kHz	**60 m UK emergency**	`SSB / CW`	UK emergency comms traffic, 60 m
3663 kHz	**80 m UK emergency**	`SSB LSB`	UK emergency comms, 80 m
7110 kHz	**40 m R1 emergency**	`SSB LSB`	IARU Region 1 emergency CoA, 40 m

FREQUENCY	NAME / CALLSIGN	MODE	WHAT IT IS
14300 kHz	**20 m global emergency**	SSB USB	Global emergency CoA, 20 m
21360 kHz	**15 m global emergency**	SSB USB	Global emergency CoA, 15 m
145.500 MHz	**2 m VHF emergency**	FM / DMR	VHF FM calling / emergency, 2 m
433.500 MHz	**70 cm VHF emergency**	FM / DMR	UHF FM calling / emergency, 70 cm

International shortwave broadcasting

Well-known international broadcasters (BBC, VOA, etc.) — these change seasonally with propagation, so treat exact kHz as a starting point. For current schedules use the BBC Global Shortwave Frequency page or Shortwave DB. The bands with the most international voice are 3.5, 7, 11, 15 and 17 MHz.

FREQUENCY	NAME / CALLSIGN	MODE	WHAT IT IS
3.5–4 MHz	**International broadcasting (AM)**	AM voice	Shortwave — many international stations
7–9 MHz	**International broadcasting (AM)**	AM voice	Shortwave — heavy international use
11–12 MHz	**International broadcasting (AM)**	AM voice	Shortwave — daytime international
15–16 MHz	**International broadcasting (AM)**	AM voice	Shortwave — daytime international
17–18 MHz	**International broadcasting (AM)**	AM voice	Shortwave — daytime international

04 Mode & Protocol Glossary

MODE	DESCRIPTION
CW	Continuous wave — Morse code (telegraphy). 200 Hz bandwidth, narrowest & most sensitive mode.
LSB	Lower Sideband SSB voice — used below 10 MHz (160/80/40 m). Dial setting = carrier, subtract ~3 kHz for tone.
USB	Upper Sideband SSB voice — used above 10 MHz (30/20/17/15/12/10 m).
AM / DSB	Amplitude Modulation, double sideband — conventional AM radio, aviation voice.
FM	Frequency Modulation — 2m/70cm/6m simplex & repeaters, commercial FM radio, 12.5 kHz channels.
J-pole	End-fed vertical: ¼λ radiator + ⅜λ folded feed stub, fed at the bottom = 11/16λ total. No ground plane needed (self-resonant), ~1 dB over a λ/4, easy to build from a single wire. The go-to portable/mobile FM vertical.

MODE	DESCRIPTION
5/8-wave	5/8λ vertical over a ground plane (radials). ~2–3 dB gain over λ/4, slightly lower feedpoint impedance — the standard for fixed VHF/UHF base stations and repeaters.
FT8	Weak-signal digital (WSJT-X) — 6-tone FSK, ~6.7 Hz bandwidth, 15 s cycles. The default HF digimode.
FT4	Faster version of FT8 — 7.5 s cycles, 4 FSK tones.
WSPR	Very weak-signal propagation (WSJT-X) — 4 FSK tones, 2.5 Hz bandwidth, 2-min cycles. Listen in the CW segment; TX anywhere in the narrowband sub-band.
PSK31	Phase-shift keying, 31 baud — keyboard data, ~31 Hz bandwidth. Popular on 40/20 m.
RTTY	Ripple Telegraphy — FSK 170 Hz shift, 45 baud. TX frequency given is the mark (audio 1700 Hz).
AMTOR / Hellschreiber	Older robust digimodes — still on the band plan, rarely used now.
DMR	Digital Mobile Radio — 2m/70cm digital voice, 4.5 kHz, BrandMeister/DMRplus repeater network.
C4FM	C4FM digital voice (Yaesu Fusion / Icom IDAS) — 6.25 kHz, 2 m.
AX.25 / APRS	Packet radio / Automatic Position Reporting — 1200 baud AFSK on 145.525 MHz (UK).
MGM	Machine-Generated Mode — umbrella for computer-driven digimodes.
CoA	Centre of Activity — recommended calling frequency for a mode/segment.
SSTV	Slow-Scan TV — image transmission, 40 m (7.175 MHz), 20 m (14.230 MHz).
DSC	Digital Selective Calling — marine digital distress/calling (Ch 16 / 85B).
ADS-B	Automatic Dependent Surveillance-Broadcast — aircraft position via Mode S 1090 MHz.

05 Antenna Guide by Band

Looking at a frequency and wondering what to build? Find the band below. The green **preferred** card is the best build for the effort — the rest are alternatives for different spaces or needs. All wire lengths are free-space values at the band centre; shorten by ≈ 5% in practice. Telescopics: set to the λ/4 length shown (with radials where noted), or full extension through an antenna tuner.

LF `30–300 kHz` λ ≈ 3,000 m at 100 kHz

Time signals (WWVB 60 kHz, WWV 100 kHz). A resonant wire here would be kilometres long — it's all about loop area or a tuner, not wavelength. Your HackRF One covers this with a loop front end.



Broadside loop PREFERRED

`1–2 m diameter (more area = more signal)`

The portable favourite. A tuned loop also rejects nearby mains/SMPS interference that drowns out LF with a whip.



Long wire + tuner

`10–25 m random wire, resonated`

Best fixed-station option. Any length works if the tuner can match it; longer and higher is better.



Telescopic + tuner

`1–2 m whip, any length, through a tuner`

Backup only. Set the whip to full extension and let the tuner do the work — expect weak signals.

Medium wave (AM radio) `526–1605 kHz` λ ≈ 300 m at 1 MHz

The classic "what antenna do I need" band — the wavelengths are far too big for a full-size wire. A loop is the portable answer; a long wire plus tuner is the fixed-station answer.



Tuned loop PREFERRED

`1.2–1.5 m diameter (8–12 turns of 1 mm wire, 5 cm core)`

Preferred for portable MW. Build a 12-turn loop on a 10 cm form with a small variable cap; it's the difference between local stations and DX.



Long wire + tuner

`15–30 m, any height`

Best fixed option. A 20 m wire strung in the garden plus a 1:99 balun and antenna tuner will pick up DX all night.



Telescopic + tuner

`1–2 m full extension + tuner`

Set to max length through a tuner — works for local stations, weak on DX. Fine for a quick check.

160 m 1.8–2.0 MHz λ/2 ≈ 79.7 m

The band where full-size antennas stop fitting. A long wire plus tuner is the standard answer; a folded/zig-zag dipole works if you have a big garden.



Long wire + tuner PREFERRED

40–80 m (even 20 m works with a good tuner)

The practical 160 m antenna. Resonate it with a tuner; it's bidirectional and forgiving about ends.

Zig-zag dipole

79.7 m total, folded to fit your space

A full λ/2 folded into a zig-zag keeps the electrical length. Needs 80 m of wire and room to snake it.



Vertical + radials

39.9 m radiator + 4 radials (≈ 10 m each)

A true 1/4-wave vertical. Radials must be near 160 m λ/4 (39 m) for best match — in practice 10 m radials + tuner.

End-fed half wave (EFHW)

79.7 m + 49:1 or 9:1 unun

One feed point at the end, convenient for roof mounting. The unun is the critical part — get a proper 49:1.

80 m 3.5–4.0 MHz λ/2 ≈ 42 m · λ/4 ≈ 21 m

A night-time DX band. Full-size dipoles are still large but manageable with a decent garden or a neighbour's trees.

Dipole PREFERRED

42 m total (21 m each side)

The classic. Hang 5–10 m up between two trees 45 m apart.



Vertical + radials

21 m radiator + 4 radials (≈ 16 m each)

Good all-round radiation. A 21 m mast with 4× 15 m radials laid on the ground works well.

End-fed half wave

42 m + 9:1 unun

Feed at one end. Works well on 80 m with a 9:1 or 49:1 unun.



1/8-wave vertical

10.5 m + ground (use a tuner)

A quarter of the space. 10.5 m over real ground is marginal but usable with a tuner — good where space is tight.

60 m 5.3–5.4 MHz λ/2 ≈ 28 m

Narrowband USB channels + WSPR. Antennas are the same as 40 m but a little longer.

Dipole PREFERRED

`28 m total (14 m each side)`

Straightforward and effective. Many 40 m dipoles work acceptably here with a tuner.



Vertical + radials

`14 m radiator + 4 radials (≈ 7 m)`

Compact option that also covers 40 m with a tuner.

40 m 7.0–7.3 MHz λ/2 ≈ 21.2 m · λ/4 ≈ 10.6 m

The best all-round band for a UK location — good DX day and night, and the antenna sizes finally fit in a garden.

Dipole PREFERRED

`21.2 m total (10.6 m each side)`

The standard build. Hang it 10–15 m up; it's the backbone of most UK home stations.



Vertical + radials

`10.6 m radiator + 4 radials (≈ 7 m each)`

The best space-saving option. A 10.6 m mast with 4× 7 m radials gives clean omni radiation.

End-fed half wave

`21.2 m + 49:1 unun`

Feed at one end — great for roof/ridge mounting with no feed line to the far end.



Half-wave vertical

`10.6 m top + 10.6 m bottom (ground-fed)`

Two 1/4-wave elements in line, fed at the junction. Higher gain than a plain vertical, needs a decent ground.

30 m 10.1–10.15 MHz λ/2 ≈ 14.8 m · λ/4 ≈ 7.4 m

The "Walfiren" band — no SSB, excellent for FT8 and DX. Antennas are small and easy.

Dipole PREFERRED

`14.8 m total (7.4 m each side)`

Fits almost any garden. A single 30 m dipole is cheap insurance for FT8 DX.



Vertical + radials

`7.4 m radiator + 4 radials (≈ 4 m)`

Compact and effective; most 40 m verticals cover 30 m too.

20 m 14.0–14.35 MHz λ/2 ≈ 10.5 m · λ/4 ≈ 5.3 m

The best value band: a 10.5 m dipole is the most useful antenna you can build for the effort — strong DX from the UK all summer.

Dipole PREFERRED

`10.5 m total (5.3 m each side)`

The one to build first. Hang 10–15 m up and it'll do 20 m properly and passably on 17/15 m.



G5RV

`10.6 m, 1:1 balanced, with a tuner`

The famous multi-band wire — works on 20 m down to 40 m with a tuner, and 15 m direct. Excellent all-rounder.



Vertical + radials

`5.3 m radiator + 4 radials (≈ 4 m)`

The smallest useful vertical. A 5.3 m pole with four 4 m radials is a weekend job.



1/8-wave vertical

`2.6 m + ground (tuner recommended)`

Tiny footprint. Usable with a tuner; fine for FT8 work in a flat.

17 m 18.06–18.16 MHz λ/2 ≈ 8.3 m · λ/4 ≈ 4.1 m

A shortwave window band — great summer DX, easy antenna sizes.

Dipole PREFERRED

`8.3 m total (4.1 m each side)`

Small and easy — fits between two garden posts.



Vertical + radials

`4.1 m radiator + 4 radials (≈ 3 m)`

A 4 m pole with radials; many 20 m verticals cover 17 m directly.

15 m 21.0–21.45 MHz λ/2 ≈ 7.0 m · λ/4 ≈ 3.5 m

The prime DX band. A 7 m dipole is one of the easiest antennas to hang and the most rewarding.

Dipole PREFERRED

`7.0 m total (3.5 m each side)`

Build a 7 m dipole — it's the best DX-per-metre of wire you can own. Hang it 10 m up.



Vertical + radials

`3.5 m radiator + 4 radials (≈ 2.5 m)`

A short mast and small radials; 15 m works great even with modest ground.

12 m 24.89–24.99 MHz λ/2 ≈ 6.0 m · λ/4 ≈ 3.0 m

A short DX band, best in summer. Same antenna logic as 15 m.

Dipole PREFERRED

`6.0 m total (3.0 m each side)`

Small dipole, easy hang. A 15 m dipole also covers 12 m with a tuner.



Vertical + radials

`3.0 m radiator + 4 radials (≈ 2 m)`

Compact; most 15 m verticals work here too.

10 m 28.0–29.7 MHz λ/2 ≈ 5.3 m · λ/4 ≈ 2.7 m

Needs sporadic-E to work DX. Antenna sizes are small; a dipole or a 2 m band vertical with a tuner covers it.

Dipole PREFERRED

`5.3 m total (2.7 m each side)`

Small and effective on E openers.



Vertical + radials

`2.7 m radiator + 4 radials (≈ 2 m)`

Most 20/15 m verticals already cover 10 m within a few metres of detune.

FM broadcast (UK radio) 87.5–108 MHz λ/2 ≈ 1.5 m at 98 MHz

For BBC Local Radio / national FM. A simple dipole or two-element Yagi is all you need; a loop is the indoor answer.



2-element Yagi PREFERRED

`Elements 1.53 m (reflector) + 1.48 m (director), boom ≈ 1 m, feed at 1.53 m element`

The outdoor winner — 4–6 dB gain over a dipole. Point it at the transmitter. Cheap to build from aluminium rod.

In-phase dipole

`1.53 m (0.77 m each side) + 4:1 balun`

Simpler than a Yagi; omni pattern. Cut 0.77 m elements, feed with a 4:1 balun or 75→50 Ω adapter.



Indoor loop

`1 m diameter, 4–6 turns`

The window-sill answer for weak indoor reception. A small ferrite or air-core loop with a variable cap.



Telescopic whip

`77–80 cm (λ/4) outdoors`

Set a 1 m telescopic to ≈ 77 cm outside a window — surprisingly good for FM with a good ground behind it.

DAB (digital FM) 174–240 MHz λ/2 ≈ 75 cm at 200 MHz

DAB/DAB+. The same antenna logic as FM but half the size.

Dipole PREFERRED

75 cm (37.5 cm each side) at 200 MHz

Simple and effective. Many car aerials and small outdoor dipoles cover the whole band well.



2-element Yagi

Elements ≈ 75 cm + 70 cm, boom 40 cm

For marginal stations — a little gain helps on the weaker multiplexes.



Telescopic whip

37–40 cm (λ/4) outdoors

Set to ≈ 40 cm outside a window; fine for strong local DAB.

6 m 50–54 MHz λ/2 ≈ 3.0 m · λ/4 ≈ 1.5 m

Amateur VHF — EME, sporadic-E and local FM. A 3 m dipole is the classic first build.

Dipole PREFERRED

3.0 m total (1.5 m each side)

The classic 6 m antenna. A 2 m dipole also works here with a tuner for SSB/CW.



J-pole

4.1 m total (1/4λ radiator + 3/8λ feed stub)

Self-resonant, fed at the bottom, no ground plane — good for rooftop FM/SSB work.



5/8-wave vertical

3.7 m + 4 radials (≈ 1.5 m)

≈ 2–3 dB over a 1/4-wave; the FM workhorse for 6 m mobile/fixed.

2 m `144–146 MHz` λ/2 ≈ 1.03 m · λ/4 ≈ 52 cm

Amateur VHF FM landmobile band. J-poles and 5/8 verticals are the standard; add a Yagi for DX.



J-pole PREFERRED

`1.4 m total`

The best first 2 m antenna — self-resonant, no radials, mounts to a single bracket.



5/8-wave vertical

`1.3 m + 4 radials (≈ 52 cm)`

2–3 dB gain over a 1/4-wave; needs the radials. The mobile classic.



3-element Yagi

`Elements ≈ 1.0/1.03/1.05 m, boom 90 cm`

The DX antenna — 6–9 dB gain. Worth it for 2 m EME/propagation work from the UK.

Dipole

`1.03 m total (52 cm each side)`

For SSB/CW on 2 m — cut 52 cm elements, feed at centre with a 4:1 balun.



Telescopic whip

`52 cm + 3–4 radials (52 cm)`

Set to 52 cm and add short ground radials; fine for local FM simplex.

70 cm `430–440 MHz` λ/2 ≈ 34 cm · λ/4 ≈ 17 cm

Amateur UHF — FM, DMR, satellite, EME. Everything is palm-sized.



J-pole PREFERRED

`47 cm total`

Best first 70 cm build — one piece of wire, no radials, mounts to a bracket.



4-element Yagi

`Elements ≈ 20–22 cm, boom 30 cm`

The satellite/EME workhorse — 10+ dB gain. A 70 cm Yagi is the single best antenna for DSAT work.



5/8-wave vertical

`43 cm + 4 radials (≈ 17 cm)`

Compact FM vertical for mobile use.



Telescopic whip

`17 cm + 3–4 radials`

Handheld territory — a 17 cm whip with radials is a decent 70 cm FM portable.

ADS-B & GPS (point frequencies) `1090 / 1575 MHz` λ/4 ≈ 7 cm (1090 MHz) · 4.8 cm (1575 MHz)

Aircraft tracking (ADS-B 1090 MHz) and GPS (L1 1575 MHz). Both are microscale — a PCB antenna is the no-compromise answer.

PCB tri-band antenna PREFERRED

`u-blox LEA/NEO style (433/915/1090 MHz)`

The standard SDR front end — a £10 PCB that handles ADS-B properly. Feed it with 50 Ω coax into your HackRF.

Short whip

`7 cm (λ/4 at 1090 MHz)`

A 7 cm wire/whip is workable for ADS-B; match it or use a stub of coax as the radiator.

GPS patch antenna

`5 cm patch over a ground plane`

For L1 1575 MHz — the ground plane is the antenna. Any 5 cm patch works.

06 How to Build Each Antenna

The guide above tells you **what** to build for each band; this section tells you **how** — materials, steps, and the mistakes that make each type fail. All lengths assume the ≈ 5% shortening; if your SWR is high, shorten the active element a few cm at a time. A **1:1 balun** is a toroid transformer that stops common-mode current on the feedline; a **4:1 unun** steps a high-impedance feed point (≈ 250 Ω) down to ≈ 50 Ω — both are 10-minute builds on a small ferrite ring, or available pre-made.

Half-wave dipole

Two equal elements totalling λ/2, fed in the middle. The reference antenna — omnidirectional, 50 Ω at the feed point, and the right first build for almost any HF band.

1. Cut two equal elements: `136 / f(MHz)` metres each — the factor already includes the ≈ 5% shortening for wire velocity and end effects. 20 m → 9.6 m each side.
2. Use 12–14 AWG copper or aluminium wire. Join the two at the centre on an insulator (wooden dowel, nylon clamp) so the elements stay parallel.
3. Connect 50 Ω coax straight to the centre join (centre conductor to one element, braid to the other) — no balun needed when it's hung in the air.
4. Hang it 5–10 m above ground, horizontal, with a few metres of clear space all round.
5. Check SWR on band and trim: high SWR → shorten, low → lengthen (a few cm at a time).

Pitfalls: The two elements must be parallel and in the same plane — a twisted join makes a figure-of-eight mess. Keep the feedline away from the elements: parallel runs couple RF onto the coax braid and distort the pattern.

Quarter-wave vertical + radials

A 1/4 λ vertical over 3–4 ground radials. Half the footprint of a dipole and easy against a wall or mast. The radials are the ground half of the antenna — without them it's a broken dipole.

1. Vertical length: `73 / f(MHz)` metres. 20 m → 3.7 m.
2. Lay 3–4 radials, each ≈ the same length as the vertical, spread at 90° to each other, as flat as you can get (0–10° from horizontal).
3. Insulate the base of the vertical (PVC sleeve or mast clip) and fix it upright.
4. Connect the coax: centre conductor to the vertical, braid to the radials.
5. If the SWR is off, add a 4:1 balun at the base.

Pitfalls: No radials = no antenna — a bare vertical is a dipole with one half missing: poor match, weak signal. More, lower and longer radials all help; a car roof or a sheet of aluminium under the base both work as ground planes.

5/8-wave vertical

A 5/8 λ vertical with 4 radials: 2–3 dB more gain and a lower angle of radiation than a 1/4-wave. The FM workhorse on 6 m / 2 m.

1. Vertical: `0.625 × 300 / f(MHz)` metres, then shorten ≈ 5%. 2 m → ≈ 1.25 m.
2. Lay 4 radials at ≈ 1/4 λ (73 / f metres), spread at 90°.
3. Fit a **4:1 unun at the base** — a 5/8-wave vertical has ≈ 250 Ω at the feed point, so a bare 50 Ω coax connection shows terrible SWR.
4. Mount, feed, and confirm resonance with an analyzer or NanoVNA.

Pitfalls: The 4:1 unun is not optional — without it the radio sees ≈ 250 Ω and most of the power reflects back. That is the #1 reason 5/8 verticals 'don't work'. A 4:1 toroid unun or the tuner's ATU both fix it.

J-pole

One continuous wire bent into a J: a 1/4 λ radiator with a stub running down one side, shorted at the bottom (total 11/16 λ). Self-resonant, no radials, no ground plane — the best first VHF antenna. The same shape in ½″ copper pipe with a T-fitting and an adjustable feed point is the classic 2 m build.

1. Cut one length: `11/16 × 300 / f(MHz)` metres (shorten ≈ 5%). 2 m → ≈ 1.35 m.
2. Bend it `3/8 × 300 / f` metres from one end — that end is the radiator (points up); the rest is the stub. 2 m: bend at ≈ 77 cm from the top.
3. Run the stub down parallel to the radiator, spaced 1–2 cm (about a wire diameter), joined at the bottom.
4. Feed the bottom join: centre conductor to the stub side, braid to the radiator side — no balun.
5. Mount on a non-conductive bracket (wood, PVC, plastic). Target SWR < 1.5:1; if high, shorten the stub a few cm at a time.

Pitfalls: Keep every metal object (bracket, mast, car body) a few cm away — a J-pole bolted to a metal roof detunes badly. If the SWR changes when you touch the coax, that's common-mode current: add a 5–10 turn ferrite choke or a 1:1 balun at the feed. Pipe builders get the same result by sliding the feed point along the stub until the NanoVNA dip lands on frequency.

Yagi

A boom carrying a reflector (behind, longest), a driven dipole (centre), and one or more directors (front, shortest). 5–10 dB of gain in one direction — the antenna for working a bearing hard, or nulled out on the noise.

1. Two-element geometry (the portability sweet spot): driven ≈ 0.47 λ, reflector ≈ 0.50 λ, spacing ≈ 0.1–0.15 λ. 2 m: driven ≈ 97 cm, reflector ≈ 103 cm, boom ≈ 30 cm.
2. Use 3/8″ aluminium tube (or thick wire) on a non-conductive boom (PVC, wood, plastic).
3. Fix the reflector and director perpendicular to the boom, insulated (PVC sleeve, nylon clip).
4. Split the driven element at its middle and feed the gap with a 1:1 balun — a split-dipole feed is ≈ 30–50 Ω and common-mode prone.
5. Point the directors at what you want to hear.

Pitfalls: The driven element must be insulated from the boom — a shorted driven element is a dead antenna. More elements = more gain but a narrower beam; 2–3 is the sweet spot for portability. On UHF a 4-element Yagi is narrow — a few degrees off-aim costs a lot of signal.

End-fed half-wave (EFHW)

A dipole with the feed point moved to one end, stepped down by a 49:1 unun. One wire, one feed point, no second element — the go-to for 'one tree, one window'. A 40 m wire also works on its harmonics up the band.

1. Wire: `468 / f(MHz)` feet — 40 m → ≈ 66 ft (20 m). 12–14 AWG insulated copper or copper-clad.
2. Wind a 49:1 unun on an FT240-43 ferrite toroid with 14 AWG enamelled wire: twist two lengths together for 2 turns (the primary), then continue the long length for 12 more turns (14 total) — 7:1 turns = 49:1.
3. Run one end of the antenna wire through the unun: centre conductor to the wire, braid to ground.
4. Run the far end of the wire up and away — as straight and high as possible.
5. Add a 1:1 coax choke (5–10 turns of coax through an FT240-31) near the feed point.
6. Tune: high SWR → shorten the wire a few cm.

Pitfalls: Miscounted turns change the ratio — count carefully (2 + 12 = 14). Without the 1:1 choke, RF runs down the coax braid and the pattern collapses. An FT240-43 handles ≈ 100 W SSB; a 100–150 pF capacitor across the unun helps the 15 m / 10 m ends.

Tuned loop (MF / AFU)

An air-core wire loop with a variable capacitor, resonated to the frequency. Narrow bandwidth (high Q) — it rejects everything off-frequency, and its figure-of-8 pattern nulls local interference. The portable answer for LF / medium wave.

1. Build a 1–1.5 m (2–4 ft) round or square frame from wood, plastic or PVC — not metal.
2. Wind 6–8 mm OFC around the frame in a box (solenoid) winding — every turn in the same plane: ≈ 8 turns on a 4 ft loop, ≈ 18 on a 2 ft loop.
3. Connect one end of the winding to a 10–365 pF AM tuning capacitor (use a 2-section capacitor for full MW coverage); the other end to ground.
4. Connect three ways: inductively (hold the receiver within 20–30 cm — the loop's field drives its ferrite), a direct tap on the winding, or a small coupling loop.
5. Tune the capacitor until the signal peaks. Rotate the loop so its face points at the station — and use the sides to null local interference.

Pitfalls: A metal frame shorts the loop out. Keep it away from mains transformers, PC power supplies and LED drivers — they drown out MW. Bigger loop = more gain; if you can't cover the whole band, the second capacitor section adds the extra capacitance. Moving the loop even a metre changes the tuning.

Long wire + tuner

Any wire, any length, matched by a tuner. The simplest 'I want to try HF' antenna: 20–50 m of wire along a roofline and an ATU.

1. Run 20–50 m of wire as high and long as you can (roofline, trees, fence).
2. Keep it clear of mains, power lines and gutters.
3. Feed with coax to an external tuner (or the radio's built-in ATU); tune for SWR < 2:1 per band.
4. Add a counterpoise (20–30 m of wire or a ground spike) at the tuner end — essential on the lower bands.

Pitfalls: Don't run the feedline parallel to the wire — it couples in RF and the pattern collapses. Without a counterpoise it's a half-wave with one half missing. Longer and higher is always better, up to a point.

Telescopic whip (portable)

The pack-and-go: a telescopic mast at 1/4 λ with 3–4 radials, or any length through a tuner. What you already carry.

1. Set the whip to 1/4 λ for VHF (FM ≈ 77 cm, 2 m = 52 cm); on HF extend it fully and use a tuner.
2. Deploy 3–4 radials: 1/4 λ if you have the space, 10–30 cm still helps a lot on VHF; 1–2 m or a ground spike on HF.
3. Connect the coax: centre to the whip, braid to the radials / ground.
4. On a car: the body IS the ground plane — clip the braid to it and skip the radials.

Pitfalls: A whip on its own is ≈ −6 dB versus a matched dipole — it works, but it's a compromise. Skipping the radials is the #1 portable mistake.

PCB / patch (UHF micro-scale)

At GPS and ADS-B frequencies (1.0–1.6 GHz) the ground plane is part of the antenna. A ≈ £10 PCB antenna is the no-compromise answer — buy it, don't build it.

1. Buy a PCB tri-band antenna (GPS / ADS-B, u-blox LEA/NEO style).
2. Feed it with a short, straight run of 50 Ω coax — don't coil it.
3. Mount on plastic or with spacers, clear of metal within a few cm.
4. Connect to the SDR; no tuning needed.

Pitfalls: Mounting a patch on bare metal makes the metal the ground plane and detunes it. Coiled feed coax couples in signal.

07 Sources

- Ofcom — UK Frequency Allocation Table (UKFAT): ofcom.org.uk (PDF, 253 pp)
- RSGB — Band Plan 2026 (effective 1 Jan 2026): rsgb.org
- Ofcom — Spectrum Information System (SIS): ofcom.org.uk
- Antenna formulas: λ/2 = c / 2f, λ/4 = c / 4f, c = 299,792,458 m/s

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